MACHINE LEARNING FOUNDATIONS (MLF)

Comprehensive Categorized Past Year Question Repository (2022 – 2025)
Word-by-Word Verbatim Extraction & Official IIT Madras Syllabus Taxonomy

Total Unique Questions: 210	**Exam Sessions Covered:** 43 Papers	**Format:** MCQ, MSQ, NAT / SA
Week 1: ML Foundations (95 Qs)	**Week 2:** Math Foundations (22 Qs)	**Week 3:** Linear Algebra (63 Qs)
Week 4: Regression & Eigenvalues (30 Qs)	**Status:** Fully Deduplicated & Verified	**Interactive Mode:** Included (JEE CBT)

■ WEEK 1 - ML FOUNDATIONS

■ Category: Data & Models (2 Questions)

MLF-Q001 | [MCQ] | Marks: +3 | *Appeared in 15 papers*

Which of these models has the smallest squared loss on this training dataset?

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

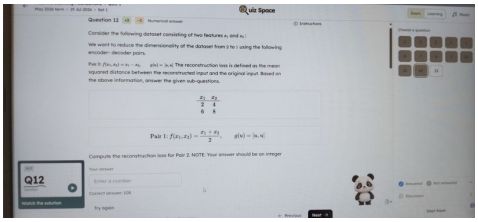
(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q002 | [SA] | Marks: +3 | *Appeared in 1 papers*

ed G) Discussion Start fresh Music Consider the following dataset consisting of two features and : We want to reduce the dimensionality of the dataset from 2 to 1 using the following encoder- decoder pairs. Pair 2: f (21 , 22) = z2, (u,u) The reconstruction loss is defined as the mean squared distance between the reconstructed input and the original input. Based on the above information, answer the given sub-questions. Pair 1: f ($1, x2) g(u) Q12 Compute the reconstruction loss for Pair 2. NOTE:



Correct Answer: 108 Watch the solution Try again Previous Next

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■ Category: ML Concepts (83 Questions)

MLF-Q003 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 2$. What is the mean squared loss of $f(x)$

(A) 0.8686

(B) 0.3636

(C) 1.0363

(D) 1.3636

MLF-Q004 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

$$Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 2

MLF-Q005 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 3.12 to 3.20

MLF-Q006 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) 2.1

(B) 2.003125

(C) 2.031

(D) 1.9968

MLF-Q007 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) 0.162

(B) 0.257

(C) 0.157

(D) 1.57

MLF-Q008 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 15

MLF-Q009 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

$$\text{Let the function } f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$$

be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

Enter the answer correct to one decimal place.

(A) (1, 3)

(B) (1/2, 1/6)

(C) (0, 0)

(D) The algorithm does not converge

MLF-Q010 | [MCQ] | Marks: +3 | *Appeared in 15 papers*

[Mathematical problem statement rendered in attached diagram below]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 72

MLF-Q011 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

(A) Yes

(B) No

MLF-Q012 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 0.25

MLF-Q013 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

P is rank one matrix.

(A) Yes

(B) No

(C) Insufficient information

MLF-Q014 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

P is rank one matrix.

(A) 3

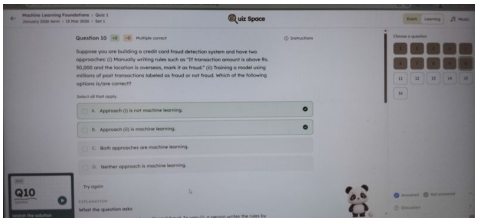
(B) 2

(C) -2

(D) 1

MLF-Q015 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

9 11 16 13 14 B, C. Approach (i) is not machine learning. Approach (ii) is machine learning* Both approaches are machine learning, Neither approach is machine learning. QIO Try again Ion What the question asks writes the rules by



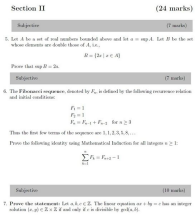
Learning Foundations > Quiz i 15 Mar 2026 > Set Question IO 42 Multiplc correct uiz Exam teatning Choose o question Suppose you are building o credit card fraud detection system and have two approaches: (i) Manually writing rules such os "If transaction amount is obove Rs. 50,000 and the location is overseas, mark it os fraud." (ii) Troining o model using millions of past transactions labeled os fraud or not fraud, Which of thc following options is/are correct? Sciect oil thot oppty. @ Instructions 9 11 16 13 14 B, C. Approach (i) is not machine learning. Approach (ii) is machine learning* Both approaches are machine learning, Neither approach is machine learning.

(A) QIO Try again Ion What the question asks writes the rules by

Explanation: *writes the rules by*

MLF-Q016 | [MCQ] | Marks: +1 | *Appeared in 3 papers*

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?


Section II (24 marks)

Subjective (7 marks)

5. Let A be a set of real numbers bounded above and let $\alpha = \sup A$. Let B be the set whose elements are double those of A, i.e.,

$$B = \{2a \mid a \in A\}$$

Prove that $\sup B = 2\alpha$.

Subjective (7 marks)

6. The **Fibonacci sequence**, denoted by F_n, is defined by the following recurrence relation and initial conditions:

$$F_1 = 1$$
$$F_2 = 1$$
$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 3$$

Thus the first few terms of the sequence are $1, 1, 2, 3, 5, 8, \ldots$.

Prove the following identity using Mathematical Induction for all integers $n \geq 1$:

$$\sum_{i=1}^{n} F_i = F_{n+2} - 1$$

Subjective (10 marks)

7. **Prove the statement:** Let $a, b, c \in \mathbb{Z}$. The linear equation $ax + by = c$ has an integer solution $(x, y) \in \mathbb{Z} \times \mathbb{Z}$ if and only if c is divisible by $\gcd(a, b)$.

MLF-Q017 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

For the input x = [0.9, 5, 1], what will be the predicted output u(z)?

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) 0

(B) 1

(C) 1.75

(D) Cannot be determined

MLF-Q018 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Match the URLs in column A with their correct rendered output in Column B with the correct app.py code for the flask application given in the previous question.

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) 1-b, 2-d, 3-a, 4-c

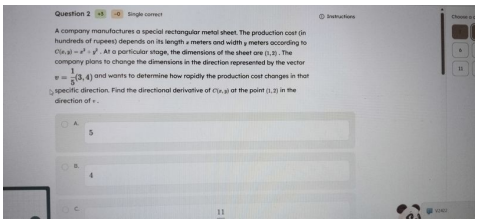
(B) 1-b, 2-a, 3-c, 4-d

(C) 1-d, 2-b, 3-a, 4- c

(D) 1-d, 2-a, 3-c, 4-b MLF

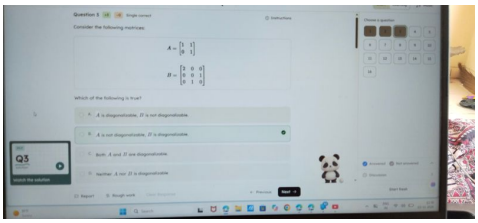
MLF-Q019 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Question 2 +3 —O Single correct A company manufactures a special rectangular metal sheet. The production cost (in hundreds of rupees) depends on its length meters and width y meters according to = + Y



MLF-Q020 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Question 3 Consider Which cf It 00? is dtoocru}iizotjie, I/ :ncl



MLF-Q021 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_112>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: <<>>

MLF-Q022 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_112>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) (A)

(B) (B) MLT

MLF-Q023 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_67>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: <<>>

MLF-Q024 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_67>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) (A)

(B) (B) PSOSM

MLF-Q025 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_83>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: <<>>

MLF-Q026 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_83>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) (A)

(B) (B) System commands

MLF-Q027 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Comprehension Data: Based on the above data, answer the given subquestions. <<<PAGE_133>>>]

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

MLF-Q028 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) (A)

(B) (B) MLT

MLF-Q029 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) (A)

(B) (B) OS

MLF-Q030 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.] Is the system Ax = b consistent?

Let the function $f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$ be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

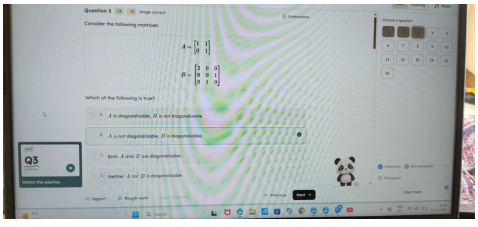
Enter the answer correct to one decimal place.

(A) Yes

(B) No

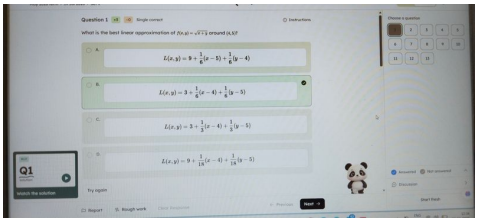
MLF-Q031 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

ed @ Discussion Start fresh ENG Music 10 15 1240 910F



MLF-Q032 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

ed C) Discussion Start fresh



MLF-Q033 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

$$Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) [1,–2, 2]

(B) [2, 2,–1]

(C) [–2, 1, 0]

(D) [–3, 0, 1]

MLF-Q034 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

$$\text{Let the function } f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$$

be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

Enter the answer correct to one decimal place.

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 5

MLF-Q035 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 1.6251 to 1.6258

MLF-Q036 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$\hat{y} = 1.34 - 0.46x$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 20.58 to 20.64

MLF-Q037 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 20.58 to 20.64

MLF-Q038 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$P \text{ is rank one matrix.}$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 1.5

MLF-Q039 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Determine which of the following limits yields a finite value.

$$P \text{ is rank one matrix.}$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

(E) [Mathematical formula option]

MLF-Q040 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Select the correct options from the following.

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q041 | [MSQ] | Marks: +2 | *Appeared in 1 papers*

Suppose you are building a credit card fraud detection system and have two approaches: (i) Manually writing rules such as "If transaction amount is above Rs. 50,000 and the location is overseas, mark it as fraud." (ii) Training a model using millions of past transactions labeled as fraud or not fraud. Which of the following options is/are correct?

$$h(x) = a + bx$$

(A) Approach (i) is not machine learning.

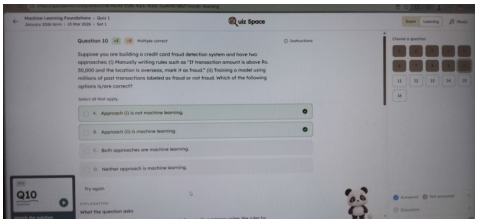
(B) Approach (ii) is machine learning.

(C) Both approaches are machine learning.

(D) Neither approach is machine learning.

MLF-Q042 | [MSQ] | Marks: +2 | *Appeared in 1 papers*

Suppose you are building a credit cord fraud detection system and hove two approaches: (i) Manually writing rules such as "'If transaction amount is above Rs. 50,000 and the location is overseas, mark it os fraud." (ii) Training o model using millions of post transactions labeled as fraud or not fraud. Which of the following options is/are correct? Select 00 thot apply. Exam Learning Choose a question 16 m Approach (i) is not machine learning. Approach (ii) is machine learning. Both approaches ore machine learninge Neither approach is machine learning, Try again QIO What the question asks



938b Hochine Learning Foundations Quiz 1 2026 term 15 Mor 2026 > Set 1 Question IO +2 -O Multiple correct Od 1 ?modez learning uiz O Instructions Suppose you are building a credit cord fraud detection system and hove two approaches: (i) Manually writing rules such as "'If transaction amount is above Rs. 50,000 and the location is overseas, mark it os fraud." (ii) Training o model using millions of post transactions labeled as fraud or not fraud. Which of the following options is/are correct? Select 00 thot apply. Exam Learning Choose a question 16 m Approach (i) is not machine learning. Approach (ii) is machine learning. Both approaches ore machine learninge

(A) Neither approach is machine learning, Try again QIO What the question asks

MLF-Q043 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following are the correct regarding the web server?

$$Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) A web server is a computer system capable of delivering web content over the internet.

(B) A web server sends the response as an HTML document which is rendered on the user's screen.

(C) A web server listens for incoming network connections on a fixed port.

(D) A web server is a software that allows to send the request over the internet. MLF

MLF-Q044 | [MSQ] | Marks: +2 | *Appeared in 3 papers*

Which of the following is/are true?

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q045 | [MSQ] | Marks: +3 | *Appeared in 6 papers*

Which of the following options is/are true?

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q046 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following statements are correct?

Let the function $f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$ be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

Enter the answer correct to one decimal place.

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q047 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following statements is/are correct?

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q048 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Which of the following statements is/are true?

$$Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) [Mathematical formula option]

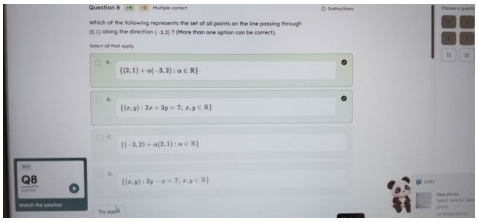
(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

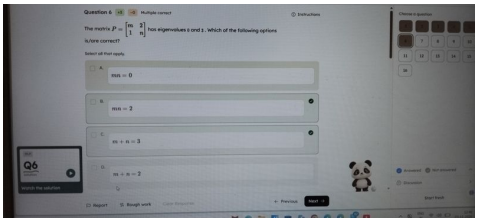
MLF-Q049 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

[j] v2422 Choose O New photo



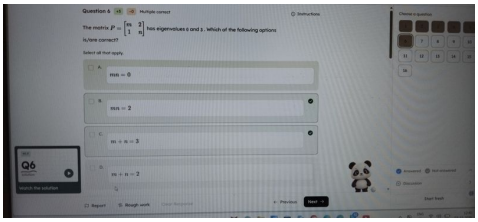
MLF-Q050 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

ed @ Discussion Start fresh % Rough work



MLF-Q051 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

ed G) Discussion Start fresh



MLF-Q052 | [SA] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Correct Answer: 458

MLF-Q053 | [SA] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)]$, $i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 484

MLF-Q054 | [SA] | Marks: +4 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 2$. What is the mean squared loss of $f(x)$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

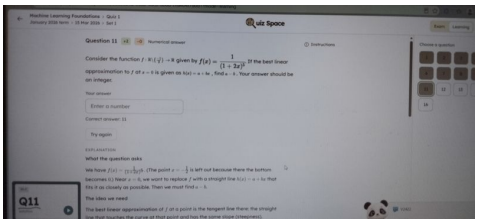
(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 2

MLF-Q055 | [SA] | Marks: +2 | *Appeared in 1 papers*

Enter o. nucnber



Correct Answer: 11 Try again What the question asks is left out because there the bottom (14205 (The point We have f(x) becomes 0) Near J (j, we wont to replace f with o straight line h(x) fits it os closely os possible. Then we tnust find The ideo we need o The best linear approximation of f ot o point is the tangent line there: the straight the curve ot that point and has the same slope (steepness). Choose a QII

Explanation: *is left out because there the bottom (14205 (The point We have f(x) becomes 0) Near J (j, we wont to replace f with o straight line h(x) fits it os closely os possible. Then we tnust find The ideo we need o The best linear approximation of f ot o poi*

MLF-Q056 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of a: (Enter answer correct to one decimal place)

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: 4

MLF-Q057 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of b: (Enter answer correct to one decimal place)

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: -3.6 to -3.4

MLF-Q058 | [SA] | Marks: +1 | *Appeared in 1 papers*

Enter the value of c: (Enter answer correct to one decimal place)

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Correct Answer: 1.45 to 1.55

MLF-Q059 | [SA] | Marks: +3 | *Appeared in 1 papers*

Find . Your answer should be an integer.

$$h(x) = a + bx$$

Correct Answer: 16

MLF-Q060 | [SA] | Marks: +3 | *Appeared in 4 papers*

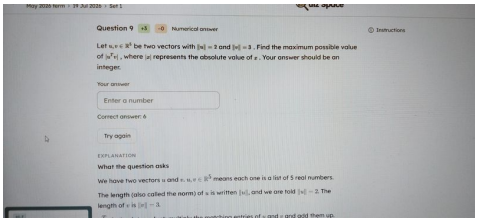
Find the value of $\alpha + 2\beta$.

$$\hat{y} = 1.34 - 0.46x$$

Correct Answer: 3

MLF-Q061 | [SA] | Marks: +3 | *Appeared in 1 papers*

Wha We Thf lie leng q est a i d liét ofS real numbers, 2. The pe are told ; ' , I k. Å I I I ' ' ' a nd add t h em u p.



Correct Answer: 6 Try again O Instructions Wha We Thf lie leng q est a i d liét ofS real numbers, 2. The pe are told ; ' , I k. Å I I I ' ' ' a nd add t h em u p.

MLF-Q062 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: <<<PAGE_52>>> Based on the above data, answer the subquestions.] Enter the value of a:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 5

MLF-Q063 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: <<<PAGE_52>>> Based on the above data, answer the subquestions.] Enter the value of b:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: -3.6 to -3.4

MLF-Q064 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: <<<PAGE_52>>> Based on the above data, answer the subquestions.] Enter the value of w:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 2.5 to 2.7

MLF-Q065 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: <<<PAGE_53>>> Based on the above data, answer the subquestions.] Enter the correct answer for d

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: 3

MLF-Q066 | [SA] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: <<<PAGE_55>>> Based on the above data, answer the subquestions.] Enter the value for m:

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: 2.8 to 3.0

MLF-Q067 | [SA] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_69>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

Correct Answer: 0

MLF-Q068 | [SA] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_84>>> Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

Correct Answer: 0

MLF-Q069 | [SA] | Marks: +3 | *Appeared in 19 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

Correct Answer: 6.37 to 6.43

MLF-Q070 | [SA] | Marks: +3 | *Appeared in 2 papers*

[Comprehension Data: Based on the above data, answer the subquestions.]

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: 0.904 to 0.910

MLF-Q071 | [SA] | Marks: +1 | *Appeared in 2 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the correct answer for a

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: 1

MLF-Q072 | [SA] | Marks: +1 | *Appeared in 2 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the correct answer for b

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Correct Answer: 2

MLF-Q073 | [SA] | Marks: +1 | *Appeared in 2 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the correct answer for c

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Correct Answer: -1

MLF-Q074 | [SA] | Marks: +1 | *Appeared in 2 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the correct answer for d

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Correct Answer: 6

MLF-Q075 | [SA] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value for c:

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
4	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Correct Answer: -1.8 to -1.6

MLF-Q076 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value of a:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 4

MLF-Q077 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value of b:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 4

MLF-Q078 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value of c:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: -4

MLF-Q079 | [SA] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value of m:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 2.8 to 3.0

MLF-Q080 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value of x:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 0.84 to 0.88

MLF-Q081 | [SA] | Marks: +1 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value of y:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x + 1$. Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$

Correct Answer: 1.71 to 1.75

MLF-Q082 | [SA] | Marks: +1 | *Appeared in 1 papers*

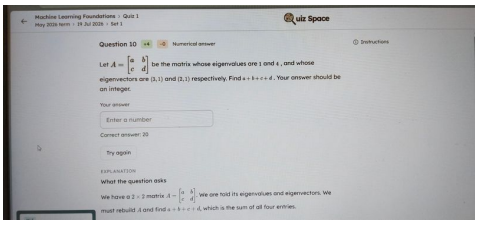
[Comprehension Data: Based on the above data, answer the subquestions.] Enter the value of z:

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x+1$. Consider the regression model $f(x) = x+1$. What is the mean squared loss of $f(x)$

Correct Answer: 0.84 to 0.88

MLF-Q083 | [SA] | Marks: +4 | *Appeared in 1 papers*

should be an integer.

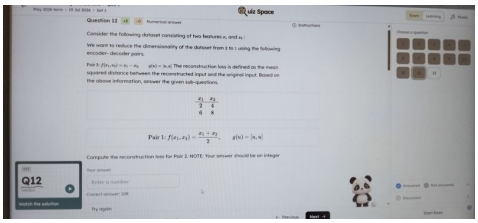


Correct Answer: 20 Try again EXPLANATION What the question asks We are told its eigenvalues and eigenvectors. We We have a 2 x 2 matrix A — must rebuild A and find a + c + d, which is the sum of all four entries.

Explanation: *What the question asks We are told its eigenvalues and eigenvectors. We We have a 2 x 2 matrix A — must rebuild A and find a + c + d, which is the sum of all four entries.*

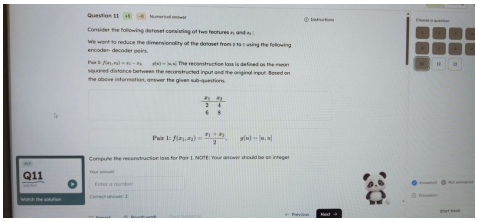
MLF-Q084 | [SA] | Marks: +3 | *Appeared in 1 papers*

the given sub-questions. Pair 1: f ($1, $2) Compute the reconstruction loss for Pair 2, NOTE:



MLF-Q085 | [SA] | Marks: +3 | *Appeared in 1 papers*

the given sub-questions. Pair 1: f (an, x2) Compute the reconstruction loss for Pair 1. NOTE:



Correct Answer: 2] 'l 13 o Answered Not answered @ Discussion Start fresh Previous Next

■ Category: ML Tasks (5 Questions)

MLF-Q086 | [MCQ] | Marks: +1 | *Appeared in 15 papers*

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

$$f(x) = 2x - 4$$

(A) Regression

(B) Classification

(C) Clustering

(D) Dimensionality Reduction

MLF-Q087 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

A hospital collects patient data including symptoms, vital signs, and lab test results. They want to find patterns to improve treatment plans. What type of task is clustering most suitable for?

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) Predicting the recovery time for a new patient.

(B) Diagnosing diseases based on labeled medical records.

(C) Grouping patients with similar symptoms and test results without prior labels.

(D) Determining the most important variable for predicting disease. Java

MLF-Q088 | [MCQ] | Marks: +2 | *Appeared in 4 papers*

Consider a dataset from an e-commerce company such as Flipkart that has details of customer behavior, which of the following problems can be best approached using clustering?

$$\hat{y} = 1.34 - 0.46x$$

(A) Predicting the exact number of sales for each customer segment.

(B) Classifying customers into segments based on labeled customer data.

(C) Identifying inherent groupings in customer data without predefined labels.

(D) Reducing the number of features in the customer dataset. Business Analytics

MLF-Q089 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

Consider the following input data points: Which among the following models has the least misclassification error?

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

(A) Sign(x1 – 2)

(B) Sign(x2 + 2)

(C) Sign(x1 + x2)

(D) Sign(x2 – x1)

MLF-Q090 | [MSQ] | Marks: +2 | *Appeared in 1 papers*

Identify which of the following requires use of classification technique.

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 2$. What is the mean squared loss of $f(x)$

(A) Predicting the amount of rainfall in May 2022 in North India based on precipitation data of the year 2021.

(B) Predicting the price of a land based on its area and distance from the market.

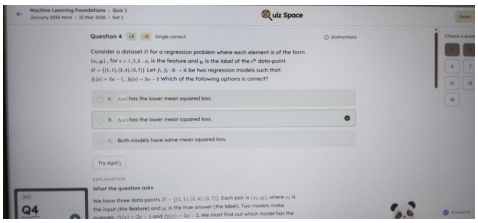
(C) Predicting whether an email is spam or not.

(D) Predicting whether the patient is having cancer or not.

■ Category: Regression Basics (2 Questions)

MLF-Q091 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

(the label). Two models make 3r '2. We must find out which model has the I and f2(J)



Mochine Learning Foundations Quiz 1 Jonuory 2026 term > 15 Mar 2026 Set 1 Question 4 -O Single correct Quiz @ Instructions Consider a dataset D for a regression problem where each element is of the form . is the feature and y, is the label of the data-point. fori I) 1), (2M), (3,7)) Let fit L : R 'Y R be two regression models such that: fl($) = 2$ 1, f2@) 3r 2 Which of the following options is correct? 11 Choose O 6 11 16 12 m (r) has the lower mean squared loss. /2($) has the lower mean squared loss. Both models have same mean squared loss. Try EXPLANATION What the question asks (3, Each pair is Yi), where is We have three data points D {(1, I), (- , the input (the feature) and yi

(A) is the true answer (the label). Two models make 3r '2. We must find out which model has the I and f2(J)

Explanation: *What the question asks (3, Each pair is Yi), where is We have three data points D {(1, I), (- , the input (the feature) and yi is the true answer (the label). Two models make 3r '2. We must find out which model has the I and f2(J)*

MLF-Q092 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Consider a dataset for a regression problem where each element is of the form , for . is the feature and is the label of the data-point. Let be two regression models such that: Which of the following options is correct?

$$h(x) = a + bx$$

(A) has the lower mean squared loss.

(B) has the lower mean squared loss.

(C) Both models have same mean squared loss.

■ Category: Types of Learning (3 Questions)

MLF-Q093 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following is/are supervised learning task(s)?

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) Predicting future sales of a product based on the past sales data.

(B) Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

(C) Recognizing and interpreting handwritten characters or words, often used in digit recognition or reading handwritten text.

(D) Categorizing text documents into predefined classes, such as spam or non- spam emails.

MLF-Q094 | [MSQ] | Marks: +2 | *Appeared in 3 papers*

Which of the following is/are supervised learning task(s)?

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

(A) Predicting future weather conditions based on the past weather observations.

(B) Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

(C) Identifying objects in images with multiple categories, such as cats, dogs, cars, and trees.

(D) Categorizing news articles into topics like politics, sports, entertainment, and technology.

MLF-Q095 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following statements is/are true?

$$Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) Email spam detection falls under classification problem.

(B) Predicting the price of a house based on previous datasets falls under classificiation problem.

(C) Classifying whether handwriting belongs to a particular person or not based on previous data falls under the classification problem.

(D) Finding out the probability that a particular poem is written by Shakespeare falls under unsupervised learning.

■ WEEK 2 - MATHEMATICAL FOUNDATIONS

■ Category: Derivatives (9 Questions)

MLF-Q096 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Consider the function : Which of the following is true?

$$h(x) = a + bx$$

(A) is continuous at , but not differentiable at .

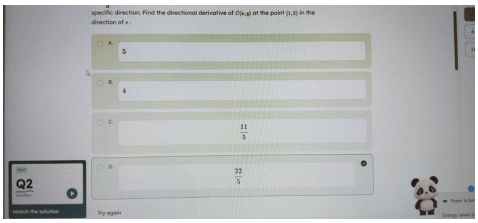
(B) is both continuous and differentiable at .

(C) is neither continuous nor differentiable at .

(D) is differentiable at , but not continuous at .

MLF-Q097 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Q2 solution Watch the solution specific direction. Find the directional derivative of at the point (1, 2) in the direction of v . . •i fbi•e•. . 11 Try again Power & Bat Energy saver is



(A) Q2 solution Watch the solution specific direction. Find the directional derivative of at the point (1, 2) in the direction of v . . •i fbi•e•. . 11 Try again Power & Bat Energy saver is

MLF-Q098 | [MCQ] | Marks: +3 | *Appeared in 4 papers*

Which of the following statements is true for a function f(x, y) that is differentiable at a point (a, b)?

$$\hat{y} = 1.34 - 0.46x$$

(A) The directional derivative exists in every direction.

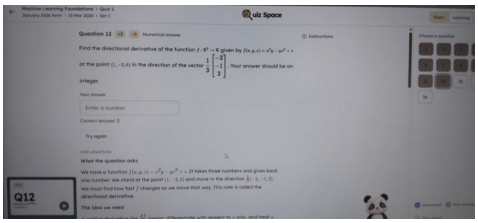
(B) The directional derivative is the same in every direction.

(C) The directional derivative can only be calculated along the axes.

(D) The directional derivative does not exist.

MLF-Q099 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

should bc on 2 Choose o 16 integers Your onswcr Enter' o nutnbcr Correct onswert 3 Try again What the question asks z) We hove a function J (x, It tokes three numbers ond qiveeo bock Q12 o 2, 0) and move in the direction one number We stand at the point (i, We must find how fost f changes as we move thot way. rote is coiled the directional derivative. The ideo we need h oect to oniv, and treat



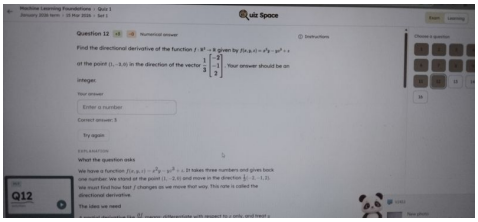
Learning Foundations > Quiz 1 term > 15 Mor 2026 Set 1 Question 12 -O Numcricot answer Find the directional derivative of the function f R' 1 at the point (It —210) in the direction of the vector — 3 Quiz Spce O Instructions R given by 2) -2 —1 Your answer should bc on 2 Choose o 16 integers Your onswcr Enter' o nutnbcr Correct onswert 3 Try again What the question asks z) We hove a function J (x, It tokes three numbers ond qiveeo bock Q12 o 2, 0) and move in the direction one number We stand at the point (i, We must find how fost f changes as we move thot

(A) way. rote is coiled the directional derivative. The ideo we need h oect to oniv, and treat

Explanation: *z) We hove a function J (x, It tokes three numbers ond qiveeo bock Q12 o 2, 0) and move in the direction one number We stand at the point (i, We must find how fost f changes as we move thot way. rote is coiled the directional derivative. The ideo we*

MLF-Q100 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

should be on 2 o Choose a 16 integer.



.Mochine teorning Foundations > Quiz term Mor 2026 Set 1 uiz Question 12 -O Numerical onswcr @ Instructions Find the directional derivative of 'the function f : R given by v,z) Av — 4 z -2 1 ot the point (1, —2,0) in the direction of thc vector —I Your answer should be on 2 o Choose a 16 integer. Your answer Enter o nurnber Correct answer: 3 Try again What the question asks We hove o function f (x, y, z) 3 Jt tokes three nurnbers ond gives bock Q12 0) ond move in the direction one number, We stand ot the point (l We must find how fast f changes as we move thot way,

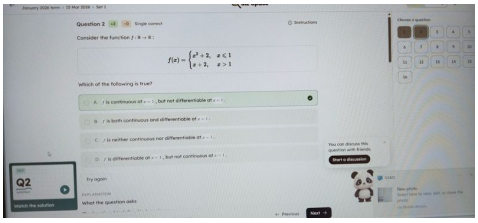
(A) This rote is coiled the directional derivative. The ideo we need ate with resoect to onlVi and treat

Correct Answer: 3 Try again What the question asks We hove o function f (x, y, z) 3 Jt tokes three nurnbers ond gives bock Q12 0) ond move in the direction one number, We stand ot the point (l We must find how fast f changes as we move thot way, This rote is coiled the directional derivative. The ideo we need ate with resoect to onlVi and treat

Explanation: *We hove o function f (x, y, z) 3 Jt tokes three nurnbers ond gives bock Q12 0) ond move in the direction one number, We stand ot the point (l We must find how fast f changes as we move thot way, This rote is coiled the directional derivative. The ide*

MLF-Q101 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

solution. Watch the solution Question 2 +2 —O Single correct Consider the function f : R R : Which of the following is true? f is continuous at 1 , but not differentiable at B. f is both continuous and differentiable at f/ is neither continuous nor differentiable at f; is differentiable at I , but not continuous at Try Ogqin o What the qUeStion asks Choose a question 14 You can discuss this question with friends. Start a discussion 11 16 v2422 New photo 12 13 15 Select here to view, 'editæ or share the photo via Mobile devices Previous Next



C) Instructions solution. Watch the solution Question 2 +2 —O Single correct Consider the function f : R R : Which of the following is true? f is continuous at 1 , but not differentiable at B. f is both continuous and differentiable at f/ is neither continuous nor differentiable at f; is differentiable at I , but not continuous at Try Ogqin o What the qUeStion asks Choose a question 14 You can discuss this question with friends. Start a discussion 11 16 v2422 New photo

(A) 12 13 15 Select here to view, 'editæ or share the photo via Mobile devices Previous Next

MLF-Q102 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

[Mathematical problem statement rendered in attached diagram below]

$$Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) The function is differentiable for all the values of x except when x = 0.

(B) The function is differentiable for all the values of x.

(C) The function is continuous at x = 0.

(D) The function is not continuous at x = 0.

MLF-Q103 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

(A) f is discontinuous at x = 0.

(B) f is differentiable.

(C) f is not differentiable at x = 0.

(D) f'(0) = 0

MLF-Q104 | [SA] | Marks: +3 | *Appeared in 1 papers*

Find the directional derivative of the function given by at the point in the direction of the vector . Your answer should be an integer.

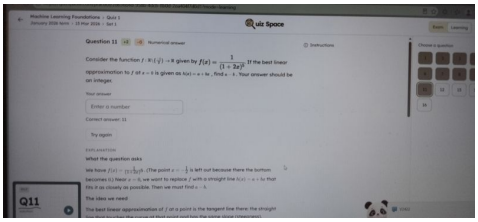
$$h(x) = a + bx$$

Correct Answer: 3

■ Category: Functions (6 Questions)

MLF-Q105 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

If the best lineor b Your onswer should be Choose a is jett out becouse there the bottom We hove f(J!) - (The point we wont 'to replace f 'With o straight line 11($) becog»es, Ot) Near titb it as closely as possible. Then we rnust find The ideo we need The best linear approximation of f at a point is the tangent line there: the straight



9e»404 f'/d0d 1 learning Hochine Learning Foundations > Quiz ZOZO ferm Is Mor 2026 set 1 Question 11 42 Consider the function f approximation to f ot on integer Your onswcr Enter o number Correct answer: 11 Try again What the question asks o Numcricol onswcr R given by f(x) 0 is given os bx , find uiz @ Instructions If the best lineor b Your onswer should be Choose a is jett out becouse there the bottom We hove f(J!) - (The point we wont 'to replace f 'With o straight line 11($) becog»es, Ot) Near titb it as closely as possible. Then we rnust find The ideo we need The best linear approximation of f at a point is the tangent line there: the straight

(A)

Correct Answer: 11 Try again What the question asks o Numcricol onswcr R given by f(x) 0 is given os bx , find uiz @ Instructions If the best lineor b Your onswer should be Choose a is jett out becouse there the bottom We hove f(J!) - (The point we wont 'to replace f 'With o straight line 11($) becog»es, Ot) Near titb it as closely as possible. Then we rnust find The ideo we need The best linear approximation of f at a point is the tangent line there: the straight

Explanation: *o Numcricol onswcr R given by f(x) 0 is given os bx , find uiz @ Instructions If the best lineor b Your onswer should be Choose a is jett out becouse there the bottom We hove f(J!) - (The point we wont 'to replace f 'With o straight line 11($) becog»*

MLF-Q106 | [MCQ] | Marks: +3 | *Appeared in 4 papers*

Which of the following functions cannot be approximated linearly around x = 0?

$$\hat{y} = 1.34 - 0.46x$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q107 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following is/are true about loss after applying Encoder and Decoder functions to a given data?

$$\mathrm{Q}_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) The loss value can be equal to any real number.

(B) The loss value cannot be negative.

(C) A perfect encoder and decoder function will not exist for any data, that is loss value can never be equal to zero.

(D) For some data, there can exist a perfect encoder and decoder functions, that is loss value can be equal to zero.

MLF-Q108 | [SA] | Marks: +1 | *Appeared in 1 papers*

Compute the equation of the function's linear approximation at point (2,3). If the equation of the linear approximation is f(x, y) = ax + by + c, then provide the values of a.

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: -2

MLF-Q109 | [SA] | Marks: +2 | *Appeared in 1 papers*

Consider the function given by . If the best linear approximation to at is given as , find . Your answer should be an integer.

$$h(x) = a + bx$$

Correct Answer: 11

MLF-Q110 | [SA] | Marks: +3 | *Appeared in 1 papers*

Give the maximum rate of change of the function at (2,3).

The eigenvalues and corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
8	$\begin{bmatrix}1\\1\end{bmatrix}$
4	$\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

Let the answer be $\begin{bmatrix}a & b\\c & d\end{bmatrix}$

Correct Answer: 2.0 to 2.5

■ Category: Metric Spaces (6 Questions)

MLF-Q111 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Let and be two lines in defined in parametric form: For example, every point on line is of the form . Which of the following is/are true?

$$h(x) = a + bx$$

(A) and are parallel

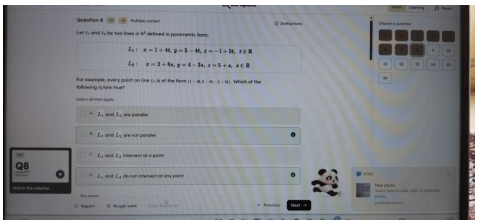
(B) and are not parallel

(C) and intersect at a point

(D) and do not intersect at any point

MLF-Q112 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Previous 11 16 New photo .12 13 14 15 Try anoin PJ Report Select here to view, edit or share the photo via Mobile devices Rough work

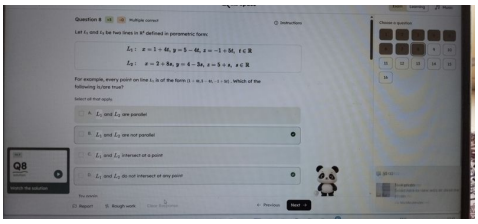


d8 Music Question 8 +3 -O Multiple correct Let LI and 142 be two lines in RS defined in parametric form: LI : c = 1 + 4t, y = 5 z c=2+8s, y=4-3s, z For example, every point on line LI is of the form (1 4 4t,5 4t, following is/are true? Select all that apply. LI and iL2 are parallel LI and 142 are not parallel LI and 142 intersect at a point n LI and do not intersect at any point Exam I Learning Choose a question 1 + 5t) . Which of the solution SVatch the solution @ Instructions Previous 11 16 New photo .12 13 14 15 Try anoin PJ Report Select here to view, edit or share the photo

(A) via Mobile devices Rough work

MLF-Q113 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Previous Music 14 11 16 12 13 Tro oonin Report % Rough work



Exam Learning Choose a question Question 8 +3 —O Multiple correct Let LI and La be two lines in R3 defined in parametric form: LI : c = 1 + 4t, y = 5 4t, z = -1+5t, teR $=2+83, z For example, every point on line LI is of the form (1 + 4t,5 —4t, 1 +5t) . Which of the following is/are true? Select all that apply. J A LI and L2 are parallel LI and L2 are not parallel C' LI and 142 intersect at a point LI and 142 do not intersect at any point O Instructions Previous Music

(A) 14 11 16 12 13 Tro oonin Report % Rough work

MLF-Q114 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following options is/are true regarding the projection matrix?

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) It is a symmetric matrix.

(B) It is an orthogonal matrix.

(C) It is orthogonally diagonalizable.

(D) It is always a rank 1 matrix.

MLF-Q115 | [MSQ] | Marks: +2 | *Appeared in 3 papers*

Which of the following options is/are true?

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) A* = A, where A* is conjugate transpose of matrix A.

(B) A is a symmetric matrix.

(C) A is a unitary matrix.

(D) Columns of A are not orthogonal.

MLF-Q116 | [MSQ] | Marks: +2 | *Appeared in 15 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.] Which of the following is/are true?

$$f(x) = 2x - 4$$

(A) A is an orthogonal matrix

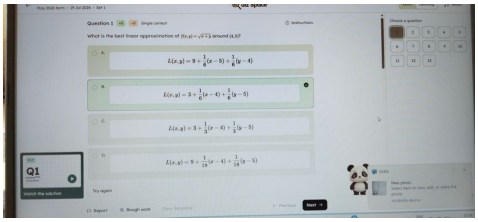
(B) A is invertible

(C) A is symmetric

■ Category: Sets & Number Systems (1 Questions)

MLF-Q117 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Next Previous Choose a question 11 V2422 New photo 12 13 18 18 QI solution Try again Watch the solution Report Select here to view, editt ot share the photo via Mobile devices Rough work



Moy 2026 term > 19 Jul 2026 Set 1 Questionl +3 —O Single correct What is the best linear approximation of = around (d, 5)? 5) O Instructions Next Previous Choose a question 11 V2422 New photo 12 13 18 18 QI solution Try

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■ WEEK 3 - LINEAR ALGEBRA & SUBSPACES

■ Category: Dot Product & Norms (2 Questions)

MLF-Q118 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

Let the vectors be orthogonal. Which of the following must be true?

$$h(x) = a + bx$$

(A) [Mathematical formula option]

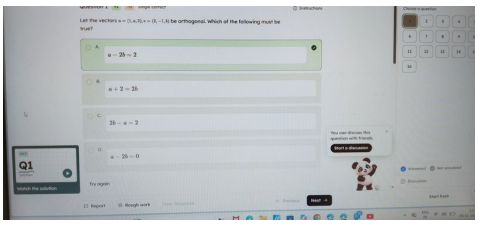
(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q119 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

ed G) Discussion Start fresh ENG



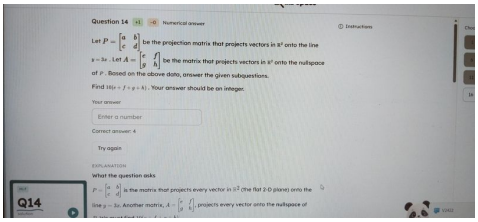
' Singte correct Let the vectors u = (1, a, = (2, —1, b) be orthogonal. Which of the following must be true? 2b 2 2b O Instructions Choose a question 11 16 12 13 14 2b QI solution, Try again Watch the solution p Report You can discuss this question with friends. Start a discussion O % Rough work Previous Next Answered Not answered G)

(A) Discussion Start fresh ENG

■ Category: Matrices (48 Questions)

MLF-Q120 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

: 4 Try again EXPLANATION What the question asks is the matrix that rojéés vecto ,inQ2kt e flat 2-D plane) onto the piojecls every vector onto the nullspace of line y —e 3:r. Another Å f



O Instructions Q14 Question 14 ricd ans Let P Y 3c . Let A of P. Base o Find loge + f 4 Your ansWer be e p oj tio be e ma he above data, n u an ers a ix at projects Oeå6rSin R2 onto the line ix hat rojects vectors iO onto the nullspace r thegi ns e Ions. o d be an Integer Choo 1b Enter a number Correct answer: 4 Try again EXPLANATION What the question asks is the matrix that rojéés vecto ,inQ2kt e flat 2-D plane) onto the piojecls every vector onto the

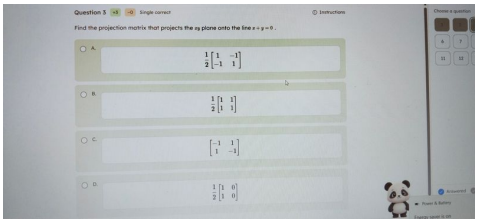
(A) nullspace of line y —e 3:r. Another Å f

Correct Answer: 4 Try again EXPLANATION What the question asks is the matrix that rojéés vecto ,inQ2kt e flat 2-D plane) onto the piojecls every vector onto the nullspace of line y —e 3:r. Another Å f

Explanation: *What the question asks is the matrix that rojéés vecto ,inQ2kt e flat 2-D plane) onto the piojecls every vector onto the nullspace of line y —e 3:r. Another Å f*

MLF-Q121 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Choose a question Find the projection matrix that projects the cy plane onto the line c + y 11 1 2 1 2 1 1 1 1 11 1 12 Answered Power & Battery Enerav saver is on



MLF-Q122 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Consider the following matrices: Which of the following is true?

$$h(x) = a + bx$$

(A) is diagonalizable, is not diagonalizable.

(B) is not diagonalizable, is diagonalizable.

(C) Both and are diagonalizable.

(D) Neither nor is diagonalizable

MLF-Q123 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

If B is diagonalizable, what is the invertible matrix P such that B = PDP–1?

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

(E) [Mathematical formula option]

MLF-Q124 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

If B is diagonalizable, which of the following is a diagonal matrix D such that B = PDP–1? Use the matrix P obtained in the previous question.

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

(E) Java

MLF-Q125 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

If the threshold in the step function is changed to z ≥ 1.0, what will happen to the misclassification rate?

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) Increase

(B) Decrease

(C) Remain same

(D) Cannot be determined Question Id : 6406531113790 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q126 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

If the threshold in the step function is changed to z ≥ 1.0, what will happen to the misclassification rate?

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) Increase

(B) Decrease

(C) Remain same

(D) Cannot be determined Question Id : 6406531114417 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q127 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

If the threshold in the step function is changed to z ≥ 1.0, what will happen to the misclassification rate?

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) Increase

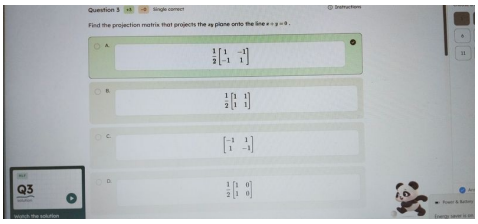
(B) Decrease

(C) Remain same

(D) Cannot be determined Question Id : 6406531114130 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q128 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Question 3 +3 —O Single correct i Find the projection matrix that projects the cy plane onto the line + y lil ill 1 1 solution Watch the solution 11 Power & Battery Energy saveris on



(A) Instructions Question 3 +3 —O Single correct i Find the projection matrix that projects the cy plane onto the line + y lil ill 1 1 solution Watch the solution 11 Power & Battery Energy saveris on

MLF-Q129 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531527853 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q130 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531527856 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q131 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531528156 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q132 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531528159 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q133 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531528818 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q134 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531528821 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q135 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531525164 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q136 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531525167 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q137 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531525791 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q138 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531525794 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q139 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531526720 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q140 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531526723 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q141 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531534289 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q142 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531534292 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q143 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531534628 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q144 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531534631 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q145 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531535065 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q146 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531535068 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q147 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531535801 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q148 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531535804 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q149 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531536141 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q150 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531536144 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q151 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531536431 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q152 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531536434 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q153 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531527044 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q154 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531527047 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q155 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531527323 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q156 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531527326 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q157 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) No

(B) Yes

(C) Insufficient information to answer the question Question Id : 6406531527602 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q158 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.]

$$f(x) = 2x - 4$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 6406531527605 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q159 | [MCQ] | Marks: +2 | *Appeared in 6 papers*

[Comprehension Data: There is no common data for this question. There are three independent sub-questions for which you have to answer True/False. <<<PAGE_87>>>] If a square matrix A is diagonalizable, then A is invertible.

$$f(x) = 2x - 4$$

(A) True

(B) False

MLF-Q160 | [MCQ] | Marks: +2 | *Appeared in 9 papers*

[Comprehension Data: There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.] If a square matrix A is diagonalizable, then A is invertible.

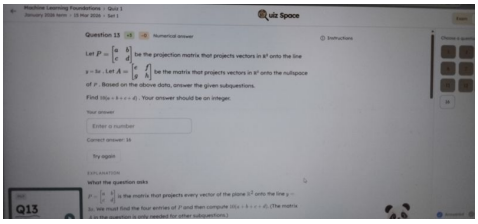
$$f(x) = 2x - 4$$

(A) True

(B) False

MLF-Q161 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

thc given Find c d) . Your onswcr should bc on inteqcr, answer Try again What the question asks onto the jtne is, rnatrix thol projects evet y vee Gf' of pione We find the four entrges of I i and 'then compule b t (The matrix the is oniv needed tor other subquestions)



Foundations Quiz Question 13 a Let P C QIS -O Numerical onvwcr uiz @ •Instructions b be thc projection matrix thot projects vector, in onto thc line d be thc motrix thot projects VCCfOt% in onto thc nu!lspocc of I' Based on thc above data, answer thc given Find c d) . Your onswcr should bc on inteqcr, answer Try again What the question asks onto the jtne is, rnatrix thol projects evet y vee Gf' of pione We find the four entrges of I i and 'then compule b t

(A) (The matrix the is oniv needed tor other subquestions)

Explanation: *onto the jtne is, rnatrix thol projects evet y vee Gf' of pione We find the four entrges of I i and 'then compule b t (The matrix the is oniv needed tor other subquestions)*

MLF-Q162 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following is/are true?

$$h(x) = a + bx$$

(A) If and are matrices of the same size, then .

(B) The rank of a matrix is equal to the number of non-zero rows in it.

(C) If is a matrix which is obtained by permuting the rows of the identity matrix, then the rank of is .

(D) If is a non-zero column vector in , then the rank of the matrix is .

MLF-Q163 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Which of the following options is/are true ?

$$\hat{y} = 1.34 - 0.46x$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 640653825040 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q164 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Which of the following options is/are true ?

$$\hat{y} = 1.34 - 0.46x$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 640653827509 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q165 | [MSQ] | Marks: +4 | *Appeared in 2 papers*

Which of the following options is/are true ?

(A) [Mathematical formula option]

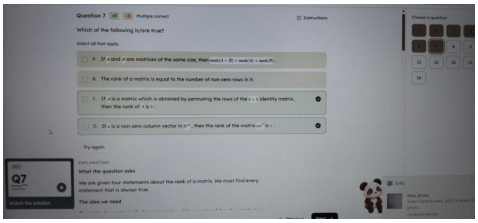
(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) Question Id : 640653827797 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

MLF-Q166 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

o Question 7 +3 -o Multiple correct Which of the following is/are true? Select all that apply. A If A and B are matrices of the same size, then rank(A+ B) — rank(A) +rank(B) B. The rank of a matrix is equal to the number of non-zero rows in it. C. If A is a matrix which is obtained by permuting the rows of the 5 x 5 identity matrix, then the rank of A is 5. D. If u is a non-zero column vector in RIO , then the rank of the matrix uuT is 1 . Try again EXPLANATION What the question asks We are given four statements about the rank of a matrix, We must find every statement that is always true. The idea we need Choose a question 11 16 IV2422 12 13

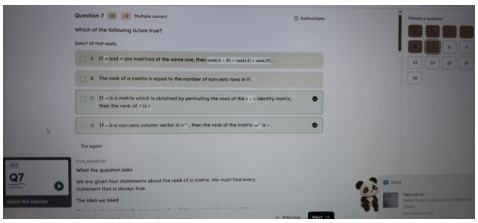


(A) @ Instructions o Question 7 +3 -o Multiple correct Which of the following is/are true? Select all that apply. A If A and B are matrices of the same size, then rank(A+ B) — rank(A) +rank(B) B. The rank of a matrix is equal to the number of non-zero rows in it. C. If A is a matrix which is obtained by permuting the rows of the 5 x 5 identity matrix, then the rank of A is 5. D. If u is a non-zero column vector in RIO , then the rank of the matrix uuT is 1 . Try again EXPLANATION What the question asks We are given four statements about the rank of a matrix, We must find every statement that is always true. The idea we need Choose a question 11 16 IV2422 12 13

Explanation: *What the question asks We are given four statements about the rank of a matrix, We must find every statement that is always true. The idea we need Choose a question 11 16 IV2422 12 13*

MLF-Q167 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

ot ion Question 7 +3 -o Multiple correct Which of the following is/are true? Select all that apply. Ifu4 and D are matrices of the same size, then rank(A + 13) — rank(A) + rank(D) . The rank of a matrix is equal to the number of non-zero rows in it. o Choose a question 9 13 14 11 16 12 C, If A is a matrix which is obtained by permuting the rows of the 5 x 5 identity matrix, then the rank of A is 5 If u is a non-zero column vector in RIO , then the rank of the matrix uuT is 1 Try again What the question asks are given four statements about the rank of a matrix. 'We must find every staferpeotjhat is always true. The idea we peed Select here to view, eci!té ot Shate •the Via, Mobife Previous Next



O Instructions ot ion Question 7 +3 -o Multiple correct Which of the following is/are true? Select all that apply. Ifu4 and D are matrices of the same size, then rank(A + 13) — rank(A) + rank(D) . The rank of a matrix is equal to the number of non-zero rows in it. o Choose a question 9 13 14 11 16 12 C, If A is a matrix which is obtained by permuting the rows of the 5 x 5 identity matrix, then the rank of A is 5 If u is a non-zero column vector in RIO , then the rank of the matrix uuT is 1 Try again What the question asks are given four statements about the rank of a matrix. 'We must find every staferpeotjhat is always true. The idea we peed Select here to view, eci!té ot Shate •the Via, Mobife Previous Next

(A)

Explanation: *are given four statements about the rank of a matrix. 'We must find every staferpeotjhat is always true. The idea we peed Select here to view, eci!té ot Shate •the Via, Mobife Previous Next*

■ Category: Span & Basis (1 Questions)

MLF-Q168 | [SA] | Marks: +4 | *Appeared in 1 papers*

Let P1 be the projection matrix onto the line spanned by the vector a = [1, 2, 3]T . Let P2 be the projection matrix that projects vectors onto the plane perpendicular to span of a. Find the determinant of P1 + P2.

$$\text{Let the function } f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$$

be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

Enter the answer correct to one decimal place.

Correct Answer: 1

■ Category: Subspaces (11 Questions)

MLF-Q169 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

Find the column space of matrix A.

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q170 | [MCQ] | Marks: +4 | *Appeared in 3 papers*

Find the row space of matrix A.

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q171 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following is a basis for the column space of the matrix given below?

$$h(x) = a + bx$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q172 | [MCQ] | Marks: +3 | *Appeared in 15 papers*

Which of the following sets forms a basis for the row space of A?

$$f(x) = 2x - 4$$

(A) {(1, 0, 2), (0, 1,–1)}

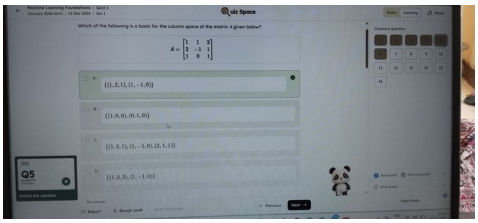
(B) {(1, 0, 2), (0, 1,–1), (0, 0,–1)}

(C) {(1, 0, 2), (0,–1, 1), (0,–1,–4)}

(D) {(0, 1,–1), (0, 0,–4)}

MLF-Q173 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

ed @ Discussion Start fresh ENG

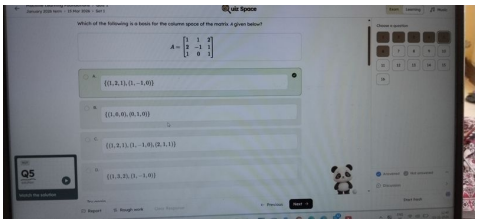


Machine Learning Foundations > Quiz 1 January 2026 term > 15 Mar 2026 Set 1 Quiz Space Which of the following is a basis for the column space of the matrix A given below? Exam Learning I Choose a question 1 1 1 o 2 1 Music 10 11 16 12 13 e , tch the solution Report {(1,3, 2), (1, % Rough work Previous Answered Not answered @

(A) Discussion Start fresh ENG

MLF-Q174 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

ed C) Discussion Start fresh Teo Report

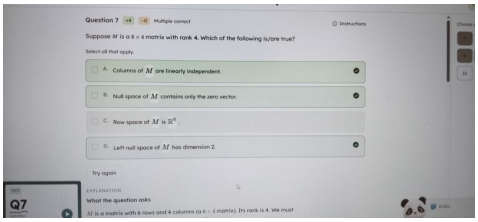


January 2026 term 15 Mar 2026 > Set 1 uiz Space Which of the following is a basis for the column space of the matrix A given below? Exam Learning Choose a question Music 14 ion Watch the solution 1 1 {(1,2, 1), (1, —1, o)) {(1,2, 1), (1, -1, o), (2, 1, 1)) % Rough work 1 o 2 1 e PrevioUS 11 16 12 13 Answered O Not answered C)

(A) Discussion Start fresh Teo Report

MLF-Q175 | [MSQ] | Marks: +4 | *Appeared in 1 papers*

Question 7 +4 -O Multiple correct Suppose M is a 6 x 4 matrix with rank 4. Which of the following is/are true? Select all that apply. A Columns of M are linearly independent. 'b Null space of M contai



MLF-Q176 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

Which of the following 3 × 3 matrices have rank 2 with column space equal to row space?

P is rank one matrix.

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q177 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following statements is/are true?

Let the function $f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$ be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

Enter the answer correct to one decimal place.

(A) Row space of a matrix is orthogonal to the null space of the matrix.

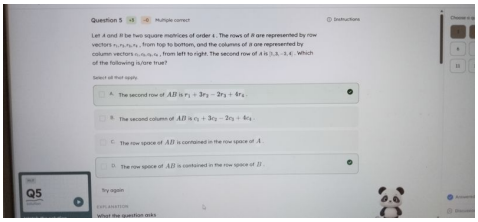
(B) Row space of a matrix is orthogonal to the null space of transpose of the matrix.

(C) Dimension of row space of a matrix is always equal to the dimension of column space of the matrix.

(D) Dimension of row space of a matrix need not be equal to the dimension of column space of the matrix.

MLF-Q178 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

o Question 5 +3 -O Multiple correct Let A and B be two square matrices of order 4 . The rows of B are represented by row vectors , from top to bottom, and the columns of B are represented by column ve



o Question 5 +3 -O Multiple correct Let A and B be two square matrices of order 4 . The rows of B are represented by row vectors , from top to bottom, and the columns of B are represented by column vectors , from left to right. The second row of A is [1, 3, —2, 4) . Which of the following is/are true? Select all that apply. m The second row of AB is r 1 4 3r2 2r3 4 B' The second column of A 13 is Cl -8- 3c2 The row space of LA 13 is contained in the row space of The row space of A 13 is contained in the row space of 13 Try again What the question asks @ Instructions Choose

(A) o

Explanation: *@ Instructions Choose o*

MLF-Q179 | [SA] | Marks: +3 | *Appeared in 1 papers*

Using the least square approximation, find the length of the projection vector p of b onto the column space of A. Enter the answer correct to two decimal places.

$$\text{Let the function } f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$$

be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

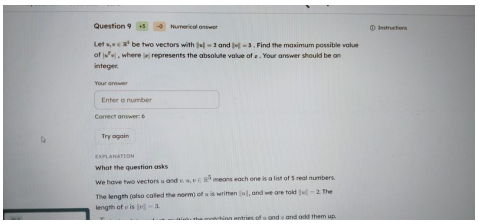
Enter the answer correct to one decimal place.

Correct Answer: 3.12 to 3.16

■ Category: Vectors (1 Questions)

MLF-Q180 | [MCQ] | Marks: +2 | *Appeared in 1 papers*

integer. Your art?wer e estion asks o vectors u, a t a so ca ed t len t ea s each oneis a list of 5 real numbers. no m) of told Ilti!1 2. The iGÅÅlGåries ! ofat and v and add them up.



■ WEEK 4 - REGRESSION & EIGENVALUES

■ Category: Eigenvalues (21 Questions)

MLF-Q181 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

(A) Eigenvectors corresponding to the two different eigenvalues of M are orthogonal.

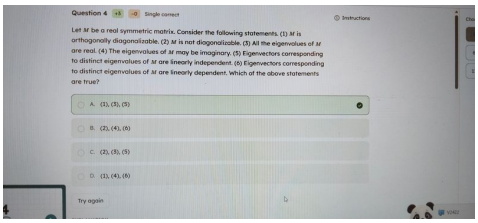
(B) The matrix M is not diagonalizable.

(C) The eigenvalues of M are 4 and 1

(D) The rank of M is 1.

MLF-Q182 | [MCQ] | Marks: +3 | *Appeared in 1 papers*

LetuVf be a reals mmetric matrix. Conside the following statements. (1) M is orthogonally diago a izable. (2) M is not di?gonalizablee (3) All$he eigenvalues of M are real. (4) The eigenvalues of M may e imaginary. (5) Eigenvectors corresponding to distinct eigenv ues of f are linearly independent. (6) Eigenvectors corresponding to distinct eige values of M are linearly; dependent. Which of the above statements are true? Try again Cha [jj V2422



MLF-Q183 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

The characteristic polynomial of a matrix with eigenvalues 1, 2, 3 and 4 is

$$\mathrm{Q}_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q184 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_112>>> Based on the above data, answer the given subquestions.] A square matrix A of shape n x n is diagonalizable if and only if it has n distinct eigenvalues.

$$f(x) = 2x - 4$$

(A) True

(B) False

MLF-Q185 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_67>>> Based on the above data, answer the given subquestions.] A square matrix A of shape n x n is diagonalizable if and only if it has n distinct eigenvalues.

$$f(x) = 2x - 4$$

(A) True

(B) False

MLF-Q186 | [MCQ] | Marks: +2 | *Appeared in 3 papers*

[Comprehension Data: <<<PAGE_83>>> Based on the above data, answer the given subquestions.] A square matrix A of shape n x n is diagonalizable if and only if it has n distinct eigenvalues.

$$f(x) = 2x - 4$$

(A) True

(B) False

MLF-Q187 | [MCQ] | Marks: +2 | *Appeared in 6 papers*

[Comprehension Data: Based on the above data, answer the given subquestions.] A square matrix A of shape n x n is diagonalizable if and only if it has n distinct eigenvalues.

$$f(x) = 2x - 4$$

(A) True

(B) False

MLF-Q188 | [MCQ] | Marks: +3 | *Appeared in 6 papers*

[Comprehension Data: There is no common data for this question. There are three independent sub-questions for which you have to answer True/False. <<<PAGE_87>>>] Which of the following is true?

$$f(x) = 2x - 4$$

(A) A does not have real eigenvalues

(B) A has two distinct real eigenvalues

A has two real eigenvalues, not necessarily distinct Question Id : 6406531525170 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern

(C) Type : NonMatrix

MLF-Q189 | [MCQ] | Marks: +1 | *Appeared in 9 papers*

[Comprehension Data: There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.] The eigenvalues of a real symmetric matrix are real.

$$f(x) = 2x - 4$$

(A) True

(B) False

MLF-Q190 | [MCQ] | Marks: +3 | *Appeared in 9 papers*

[Comprehension Data: There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.] Which of the following is true?

$$f(x) = 2x - 4$$

(A) A does not have real eigenvalues

(B) A has two distinct real eigenvalues

A has two real eigenvalues, not necessarily distinct Question Id : 6406531527859 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern

(C) Type : NonMatrix

MLF-Q191 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) Eigenvalues of A are 1, 2, and 3.

(B) Eigenvalues of A are 1, 2, and 2.

(C) Eigenvalues of B are 2, 20, and 78.

(D) Matrix B is diagonalizable.

MLF-Q192 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

P is rank one matrix.

(A) B has eigenvalues -4, 8, 70.

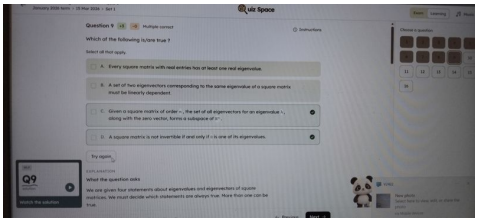
(B) A has eigenvalues 1, 16, 81.

(C) B is an upper triangular matrix.

(D) B is a lower triangular matrix.

MLF-Q193 | [MSQ] | Marks: +2 | *Appeared in 1 papers*

Exom j Leorning Choose a question 11 16 12 13 solution. Every square matrix with real entries has at least one real eigenvalue. A set of two eigenvectors corresponding to the same eigenvalue of a square matrix must be linearly dependent. Given a square matrix of order n j the set of all eigenvectors for an eigenvalue A , along with the zero vector, forms a subspace of RO . tx A square matrix is not invertible if and only if 0 is one of its eigenvalues. Try again Woot the question asks We *four statements about eigenvalues and eigenvectors of square matrices) We must decide which statements are always true. More than one can be true, 14 Select here to 'View, edit, or share the



Jonuory 2026 term > 15 Mor 2026 Set Question 9 -O Multiple correct Which of the following is/are true ? Select 011 that applys uiz Space @ Instructions Exom j Leorning Choose a question 11 16 12 13 solution. Every square matrix with real entries has at least one real eigenvalue. A set of two eigenvectors corresponding to the same eigenvalue of a square matrix must be linearly dependent. Given a square matrix of order n j the set of all eigenvectors for an eigenvalue A , along with the zero vector, forms a subspace of RO . tx A square matrix is not invertible if and only if 0 is one of its eigenvalues. Try again Woot the question asks We *four statements about eigenvalues and eigenvectors of square matrices) We must decide which statements are always true. More than one can be true, 14

(A) Select here to 'View, edit, or share the

MLF-Q194 | [MSQ] | Marks: +4 | *Appeared in 3 papers*

Let A and B be symmetric matrices such that AB = BA. Then which of the following options is/are true?

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) AB is a symmetric matrix.

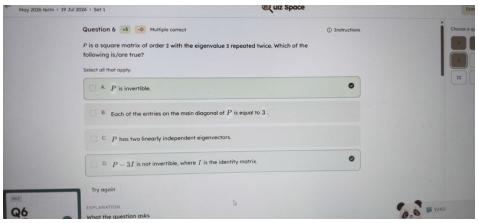
(B) AB is orthogonally diagonalizable.

(C) BA has real eigenvalues.

(D) AB is not a symmetric matrix.

MLF-Q195 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

P is a square matrix of order 2 with the eigenvalue 3 repeated twice. Which of the following is/are true? Select all that apply. P is invertible. Each of the entries on the main diagonal of P is equal to 3 p has two linearly independent eigenvectors. D. p 31 is not invertible, where I is the identity matrix. Try again EXPLANATION hat the question asks Exar Choose a 11



May 2026 term > 19 Jul 2026 > Set 1 Question 6 +3 -O Multiple correct ur space O Instructions P is a square matrix of order 2 with the eigenvalue 3 repeated twice. Which of the following is/are true? Select all that apply. P is invertible. Each of the entries on the main diagonal of P is equal to 3 p has two linearly independent eigenvectors. D. p 31 is not invertible, where I is the identity matrix. Try again EXPLANATION hat the question asks Exar Choose a 11

(A) a 11

Explanation: *hat the question asks Exar Choose a 11*

MLF-Q196 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

The matrix has eigenvalues and . Which of the following options is/are correct?

$$h(x) = a + bx$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q197 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

Which of the following is/are true ?

$$h(x) = a + bx$$

(A) Every square matrix with real entries has at least one real eigenvalue.

(B) A set of two eigenvectors corresponding to the same eigenvalue of a square matrix must be linearly dependent.

(C) Given a square matrix of order , the set of all eigenvectors for an eigenvalue , along with the zero vector, forms a subspace of .

(D) A square matrix is not invertible if and only if is one of its eigenvalues.

MLF-Q198 | [MSQ] | Marks: +4 | *Appeared in 4 papers*

Which of the following options is/are true?

$$\hat{y} = 1.34 - 0.46x$$

(A) A is a symmetric matrix.

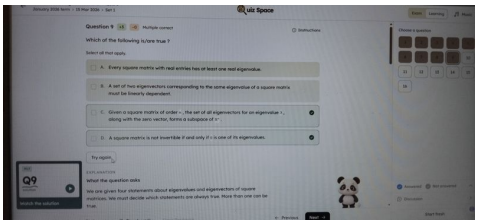
(B) A is a diagonal matrix.

(C) A is orthogonally diagonalizable.

(D) Eigenvalues of A are -1, -1 and 2.

MLF-Q199 | [MSQ] | Marks: +3 | *Appeared in 1 papers*

ed C) Discussion Start fresh Next



January 2026 term 15 Mar 2026 > Set 1 Question 9 +3 —O Multiple correct Which of the following is/are true ? Select all that apply. uiz Space @ Instructions Exam Learning Choose a question Music 11 16 12 the' solution AD Every square matrix with real entries has at least one real eigenvalue. ulo BQ A set of two eigenvectors corresponding to the same eigenvalue of a square matrix must be linearly dependent. C. Given a square matrix of order n , the set of all eigenvectors for an eigenvalue , along with the zero vector, forms a subspace of e A square matrix is not invertible if and only if o is one of its eigenvalues. Try agam EXPLANATION What the question asks We are given foUrstatements about eigenvalues and eigenvectors of square matricefr'l We must decide which statements are always true. More than one can be true, Previous Answered Not answered C) Discussion Start fresh

(A) Next

Explanation: *What the question asks We are given foUrstatements about eigenvalues and eigenvectors of square matricefr'l We must decide which statements are always true. More than one can be true, Previous Answered Not answered C) Discussion Start fresh Next*

MLF-Q200 | [SA] | Marks: +2 | *Appeared in 1 papers*

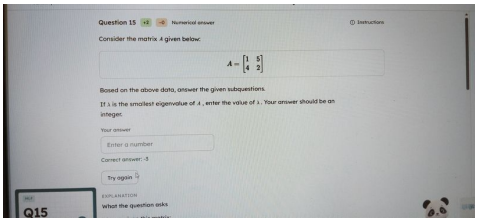
If is the smallest eigenvalue of , enter the value of . Your answer should be an integer.

$$h(x) = a + bx$$

Correct Answer: -3

MLF-Q201 | [SA] | Marks: +2 | *Appeared in 1 papers*

the givensu questions. If is the smallest eigenvalue ofu4 , enter the value of X .



Correct Answer: -3 Try again EX PLAN ATION: What the question asks Q15
Explanation: *Q15*

■ Category: Eigenvectors (3 Questions)

MLF-Q202 | [SA] | Marks: +2 | *Appeared in 1 papers*

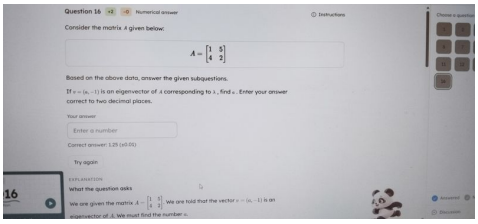
If is an eigenvector of corresponding to , find . Enter your answer correct to two decimal places.

$$h(x) = a + bx$$

Correct Answer: 1.24 to 1.26

MLF-Q203 | [SA] | Marks: +2 | *Appeared in 1 papers*

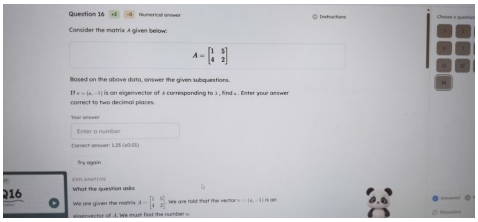
the given subquestions. (a, —1) is an eigenvector of A corresponding to , find a . Enter your answer correct to two decimal places.



Correct Answer: 1.25 (±O.OI) Try again EXPLANATION What the question asks o We are told that the vector We are given the matrix Ä eigenvector of A. We must find the number a. a, 1) is an Choose a question 16 Answered ft @ Discussion
Explanation: *What the question asks o We are told that the vector We are given the matrix Ä eigenvector of A. We must find the number a. a, 1) is an Choose a question 16 Answered ft @ Discussion*

MLF-Q204 | [SA] | Marks: +2 | *Appeared in 1 papers*

the given subquestions. (a, —1) is an eigenvector of A corresponding to A , find a . Enter your answer correct to two decimal places.



Correct Answer: 1.25 (±O.OI) Try again EXPLANATION What the question asks We are told that the vector v We are given the matrix A eiaenvector of We must find the number a. l) is an Choose a question Answered C) Discussion

Explanation: *What the question asks We are told that the vector v We are given the matrix A eiaenvector of We must find the number a. l) is an Choose a question Answered C) Discussion*

■ Category: Error Calculations (2 Questions)

MLF-Q205 | [MSQ] | Marks: +3 | *Appeared in 3 papers*

[Mathematical problem statement rendered in attached diagram below]

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

(A) Square error loss ≤ 0-1 loss

(B) Square error loss ≥ 0-1 loss

(C) The 0-1 loss is 0.25

(D) The square error loss is 1.25

MLF-Q206 | [SA] | Marks: +4 | *Appeared in 1 papers*

Give the reconstruction error for this encoder decoder pair. The reconstruction error is the mean of the squared distance between the reconstructed input and input.

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 2$. What is the mean squared loss of $f(x)$

Correct Answer: 24 to 25

■ Category: Least Squares (4 Questions)

MLF-Q207 | [MCQ] | Marks: +4 | *Appeared in 3 papers*

Find the best fit line for the dataset given below using least squares method.

x	0	2	3	6	8
y	-2	5	3	1	-3

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q208 | [MCQ] | Marks: +4 | *Appeared in 1 papers*

For the above given data set Raj wants to compute a least square approximation that best fits the dataset with minimum error. Which of the following is the correct equation for the best fit line computed by Raj?

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 2$. What is the mean squared loss of $f(x)$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

(E) Java

MLF-Q209 | [MCQ] | Marks: +3 | *Appeared in 3 papers*

[Comprehension Data: Based on the above data, answer the given subquestions. <<<PAGE_133>>>] Find the projection of b onto U.

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

(A) [Mathematical formula option]

(B) [Mathematical formula option]

(C) [Mathematical formula option]

(D) [Mathematical formula option]

MLF-Q210 | [SA] | Marks: +1 | *Appeared in 3 papers*

Find the length of the error vector e =b – p, where p is the projection of b onto U. Enter the answer correct to two decimal places.

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

Correct Answer: 3.70 to 3.78